

Technology-wise, it is OLED, solar and wireless LED

Here are some technological trends emerging in the LED space:

Colour-changing LEDs. "One key trend is the development of LED lights that can produce any light within the visible spectrum. Philips is the first to introduce such LED bulbs, which can change the colour of light," notes Raushan. With the help of an app, the consumer can control the colour of LED.

Wireless-bonded LEDs. Development of wireless-bonded LED technology can be considered to be another latest technology trend with regard to LED lighting market. Raushan says, "Wireless-bonded LED technology is also referred as flip-chip. It provides benefits such as improved durability, better heat dissipation and longer life."

Wireless LEDs. Modern buildings being constructed today comprise lighting arrangements with daylight, occupancy and time scheduling on/off sensors. "Similarly, many upcoming LED road lighting systems call for wireless operability, daylight and on/off scheduling," says Lalchandani. The new LED lighting systems (both for indoor and street-lighting) need to be designed to seamlessly integrate and work with these sensors and control systems.

OLEDs. There is a new trend of solar LED lights being used to provide low-cost lighting in rural households. "The newest technology trend observed in the LED market is the organic light emitting diode (OLED) technology. OLEDs are LEDs manufactured from polymers with a liquid crystal display (LCD) glass covered with indium tin oxide (ITO) and polymer materials applied by ink jet printing," informs Lahiri.

absence of toxic materials, are other major driving factors that make LEDs more affordable and growth-driving, informs Harmeet Singh, technical lead, Analog Applications, TI India.

"The average lumen efficiency is much higher for LEDs when compared to incandescent lamps and CFL lamps," notes Swetanta Lahiri, research analyst, Netscribes. Energy conservation is driving demand in the LED sector. With LEDs, you can save approximately 80 per cent of energy as compared to normal lighting devices.

LEDs have a longer life in comparison to normal lighting devices or even CFLs. Dhawan informs, "In general, life of an LED is approximately 25,000 burning hours, whereas ordinary lights last around 1000 burning hours and CFLs around 5000 to 6000 burning hours. Consumer's demand is driven by the fact that once he or she has used LEDs, extra costs for maintenance or replacement will not be incurred for a long period of time."

Awareness with regard to energy-saving has a huge impact on the development of energy-efficient lighting systems such as LED light bulbs. "People are now more inclined toward buying LED light bulbs instead of CFLs or other inefficient light bulbs. This is further being supported by the

declining prices of LED light bulbs," says Raushan.

Government policies are helping

The Indian market for LEDs has shown a steady growth and increasing demand over the last couple of years, thanks to the initiatives taken by the government. "Newly-formed government in India has emphasised that each and every house should have a light bulb by 2019, creating huge demand for power by 2019. This has led to the requirement of energy-efficient lighting systems such as LED light bulbs," says Raushan.

Street-lighting, and industrial and commercial applications are boosting the market in India. Government has also emphasised that all street lights in public places should be replaced with LED light bulbs, encouraging the adoption of LED lighting in India. Street-lighting has huge scope for energy-saving that can be achieved with the implementation of LED light bulbs.

The largest segment driving LED adoption is the street-lighting segment, informs Harish Lalchandani, country head, GE Lighting. He says, "In a bid to cut energy costs and increase the quality of light toward enhancing town and pedestrian safety, the government

is pushing conversion of all street lights to LED. According to ELCOMA reports, there are about 27.5 million street-lighting points in India that will be converted to LED in a phased manner."

Government has taken several other steps to ensure the implementation of LED at all levels. "Recently, the government of Andhra Pradesh has announced that it will be distributing LED light bulbs to 3.7 million households in Andhra Pradesh at a cost of ₹ 10 (US\$ 0.16 approx.), which usually cost approximately \$6," informs Raushan.

LED standards were published in 2012, which brought in a large-scale lighting revolution in India, shares Lahiri. He says, "Customs duty on LED was reduced from 10 per cent to 4 per cent as part of government of India initiatives."

Government is also kick-starting demand in the domestic segment, notes Dhawan. He says, "Government, through its energy-development agencies in various states, is promoting a lot of projects where they are going to replace the existing lighting systems with LED. In the last one year or so, I have seen a lot of such projects coming up and agencies being set-up by the government."

Power Ministry has also intended to provide LED technology under Rajiv Gandhi Grameen Vidyutikaran Yojana (RGGVY) initiative. Raushan says, "Government has intended to replace all street-lights and lights in public places with LED light bulbs. It has modified all CFL distribution scheme into LED distribution. All government departments have been asked to procure only LED light bulbs instead of incandescent bulb and CFL light bulbs." He adds, "All this development is expected to drive the demand, as well as awareness, amongst the consumer regarding LED light bulbs."

Talking about how the Ministry of Power played an important role in promoting the use of LED lamps under RGGVY scheme, Lahiri says, "Railways have already started using LEDs in different utilities, such as traffic signals and railway coaches."

Finally, government of India initiatives like 'Bachat Lamp Yojna' and 'LED